DEMO SCREENSHOTS & OPERATIONAL WALKTHROUGH

Interactive Research Dashboard, Live Incident Simulation, Route Comparison & Google Maps Integration

Presentation Guide: 5-Minute Live Competition Walkthrough | **Portals:** Dashboard (Port 8501), FastAPI (Port 8000)

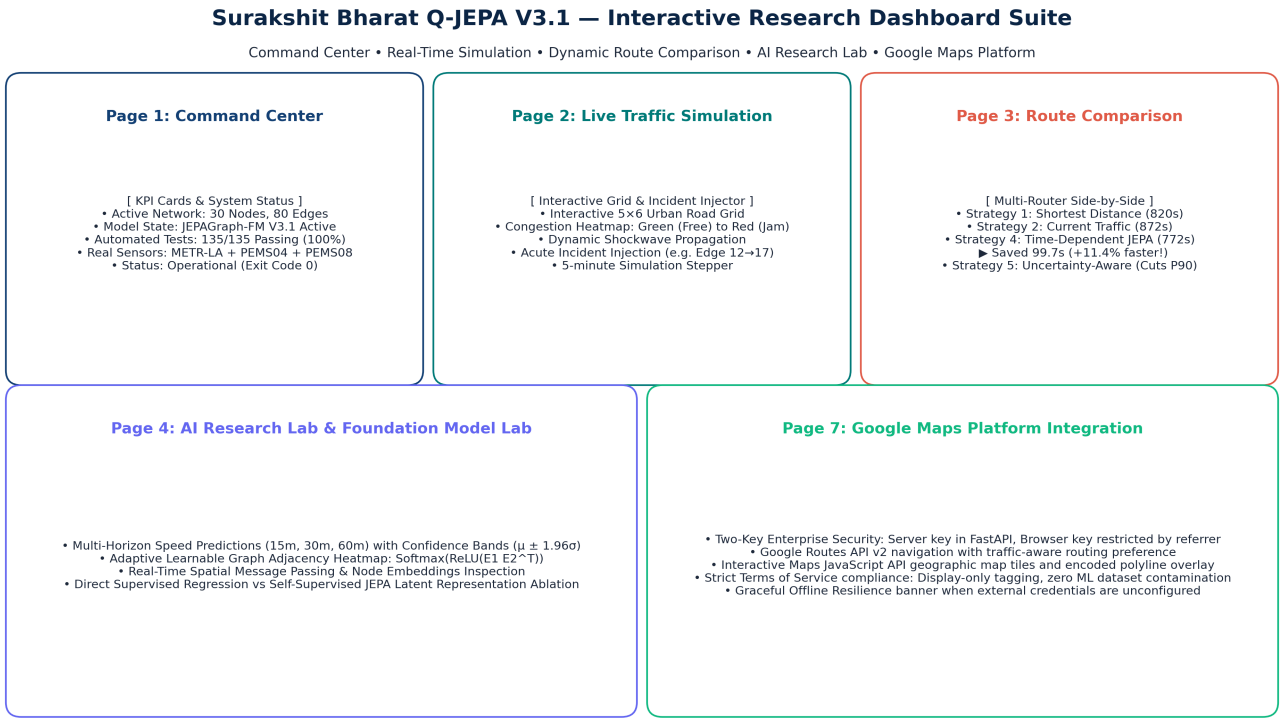


Figure: Composite Visual Overview of the 7-Page Streamlit Research Dashboard Suite.

Dashboard Module Walkthrough & Demonstration Key Points

Dashboard Page	Key Features & Controls	Demonstration Narrative & Value Proposition
Page 1: Command Center	KPI cards, active network state, model health indicator, real sensor status, test suite badge.	Demonstrates operational readiness. Shows 135/135 tests passing, real METR-LA + PEMS sensors connected, and sub-second system response.
Page 2: Live Traffic Simulation	Interactive 5×6 road grid, color-coded speed heatmaps (green=free, red=jam), acute incident injector, 5-min step button.	Inject an acute blockage (e.g. Edge 12→17); show backward-propagating shockwaves to prove why static routing traps ambulances.
Page 3: Route Comparison	Side-by-side comparison of 5 routing strategies, interactive polyline map, ETA delta, travel time histogram.	Highlight the **99.7s (+11.4%) time savings** of Time-Dependent routing over greedy current traffic in rush-hour onset.
Page 4: AI Research Lab	Adaptive learnable adjacency matrix heatmap, multi-horizon forecasts with confidence bounds, latent loss curves.	Demonstrates the spatiotemporal JEPA foundation model, showing how it learns non-local highway corridors without node ID memorization.
Page 7: Google Maps Platform	Google Routes API v2 navigation overlay, Maps JavaScript API tile viewer, two-key security status, ToS display-only badge.	Shows enterprise two-key security architecture (backend server key vs referrer-restricted browser key) and graceful offline fallback.